

Python Programming

Overview

Python is a high-level, interpreted programming language known for its simplicity and readability. Created by Guido van Rossum in 1991, it has become one of the most popular languages worldwide, used in web development, data science, automation, AI, and more.

Key Features

- Simple and readable syntax — code reads almost like English.
- Dynamically typed — no need to declare variable types.
- Huge standard library and ecosystem (PyPI has 400,000+ packages).
- Supports multiple paradigms: OOP, functional, procedural.
- Cross-platform: runs on Windows, macOS, Linux.

Basic Example

```
# Hello World and basic operations
print('Hello, World!')

name = 'Manoj'
age = 27
print(f'My name is {name} and I am {age} years old.')
```

Core Concepts

Python programs are built around variables, loops, functions, and classes. Everything in Python is an object, making it highly flexible and consistent.

Functions and Classes

```
def greet(name):  
    return f'Hello, {name}!'  
  
class Person:  
    def __init__(self, name, age):  
        self.name = name  
        self.age = age  
  
    def info(self):  
        return f'{self.name}, age {self.age}'  
  
p = Person('Manoj', 27)  
print(p.info())
```

Common Use Cases

- Web development using Django or FastAPI.
- Data analysis with Pandas and NumPy.
- Machine learning with TensorFlow and PyTorch.
- Test automation using Pytest, Selenium, or Playwright.
- Scripting and DevOps automation.